

DTSA 5748

Deep Learning for Natural Language Processing

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (81.5 / 100)	60.2 / 100
SPECIALIZATION Natural Language Processing		

60

CURRENT AVG SCORE
across 6 job roles

92

POTENTIAL SCORE
with recommended additions

+46

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Deep learning has revolutionized the field of natural language processing and led to many state-of-the-art results. This course introduces students to neural network models and training algorithms frequently used in natural language processing. At the end of this course, learners will be able to explain and implement feedforward networks, recurrent neural networks, convolutional neural networks, and transformers. They will also have an understanding of transfer learning, the paradigm behind popular models such as BERT and GPT-3.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	81.5	100.0	+18.5	MEDIUM
Machine Learning Engineer	71.3	100.0	+28.7	MEDIUM
Data Scientist	63.6	92.1	+28.5	MEDIUM
Applied Scientist	53.5	100.0	+46.5	HIGH
ML Engineer	53.5	87.7	+34.2	HIGH
MLOps Engineer	37.7	71.2	+33.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Deep Learning 1 of 6 roles

cnn

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: cnn.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

81.5 → 100.0 (+18.5 pts)

data build technical systems problems technology detection machine

Machine Learning Engineer

71.3 → 100.0 (+28.7 pts)

data analytics business mobile design science applications machine

Data Scientist

63.6 → 92.1 (+28.5 pts)

data analytics simply mitigation social build problems approaches

Applied Scientist

53.5 → 100.0 (+46.5 pts)

lab discovery scientific research literature problems technical data

ML Engineer

53.5 → 87.7 (+34.2 pts)

cnn discovery data machine systems build problems impact

MLOps Engineer

37.7 → 71.2 (+33.5 pts)

detection data text systems machine image error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.